

AI FOR VALIDATION

CHAPTER 1

From Model Validation to Agentic Model Validation

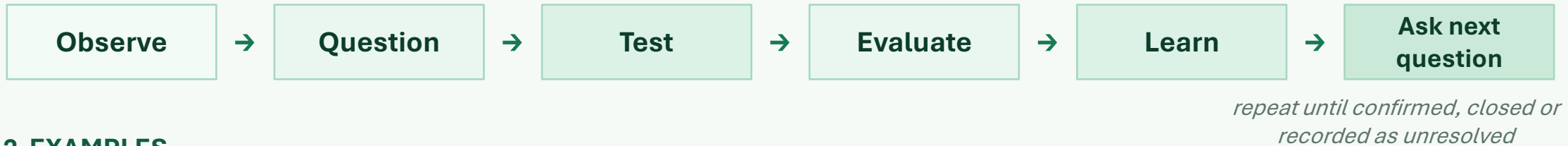
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Model Validation as a Discovery Loop

Validation is an investigation: each result determines the next question.

1 COMMON LOOP



2 EXAMPLES

Conceptual Soundness

1. Start with development data, intended use and the population the model is supposed to serve.
2. Examine input–outcome relationships: shape, thresholds, nonlinearities and interactions.
3. Compare empirical relationships with fitted model behavior; test instability across populations and vintages.

Weakness Identification

1. Begin with candidate failure conditions, then execute and measure them.
2. Compare competing weaknesses and concentrate testing on promising regions or conditions.
3. Generate refined failure conditions, repeat the search and independently confirm selected weaknesses.

3 KEY PRINCIPLE

Validation is adaptive and connected: evidence from one test determines the next question, can trigger another validation activity and complements, rather than replaces, required validation work.

Validation as an Agentic System

The Discovery Loop becomes an executable, adaptive and governed system.

1

Discovery Loop becomes the agentic engine

The system observes evidence, asks the next question, runs the test, evaluates the result and continues the investigation.

2

AI moves from assistance to deeper validation

AI is no longer just summarizing, drafting or speeding up tasks; it helps pursue issues and interactions a fixed plan may miss.

3

Governance is built into the system

The agent can propose and investigate, but verification and final authority remain separate: Proposer $\neq$ Verifier $\neq$ Authority.



Discovery loop as executable workflow